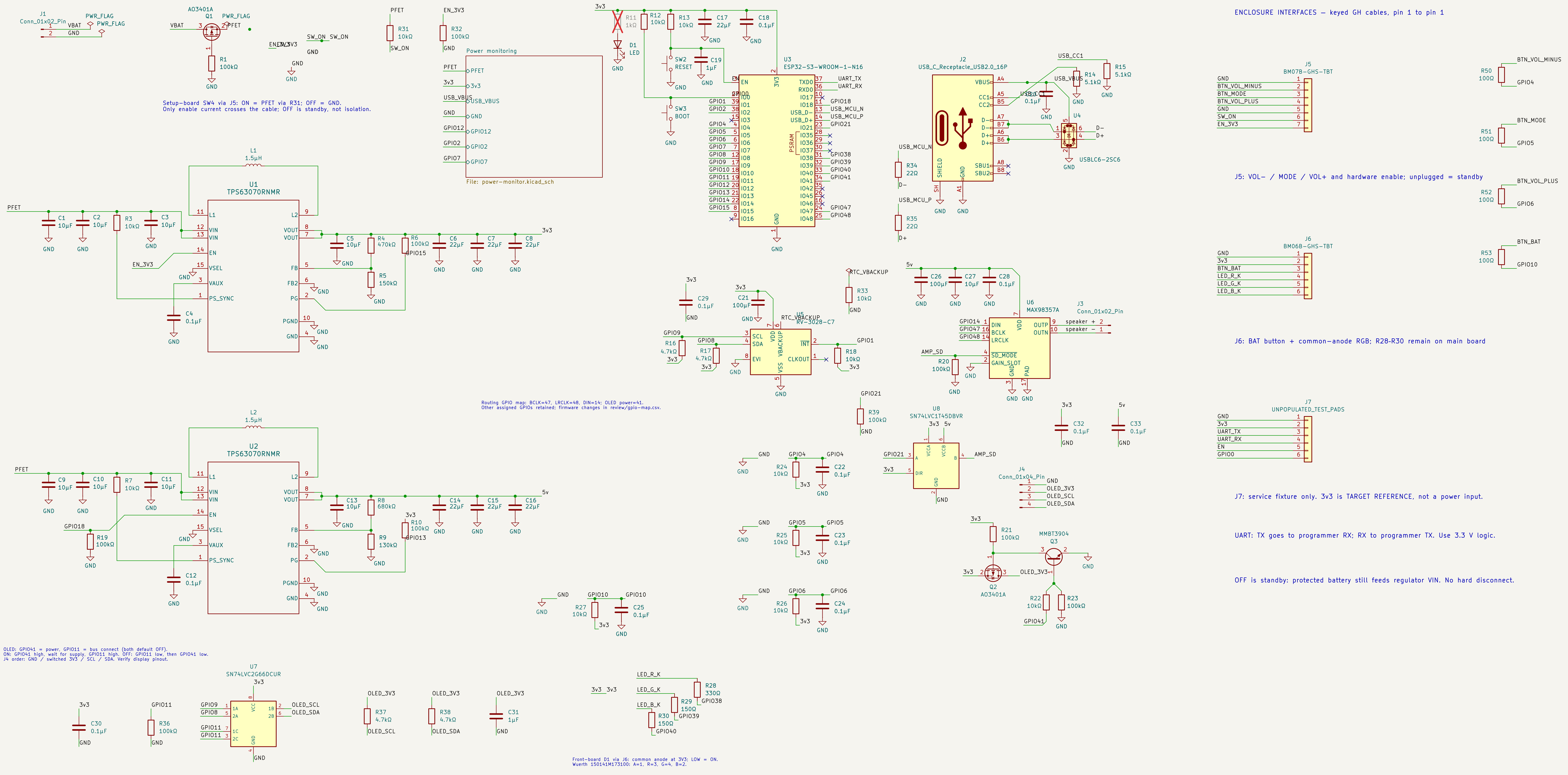
DESIGN REVIEW 2026-09-04 | Prototype; battery/load and enclosure pending confirmation.
TPS63070: PS/SYNC HIGH = power save (both rails). 3.31V / 4.985V nominal.
Cold start requires >=3.0V at converter input. JST-PH input rated 2A with specified wire.
USB is data-only: VBUS supplies ESD bias only. R11 DNP for low standby current.
RTC backup/trickle charge OFF; time is lost when main power is off.
Capacitance must be checked after DC bias; inspect review/design-review.md before fabrication.



Setup-board SW4 via J5: ON = PFET via R31; OFF = GND.
Only enable current crosses the cable; OFF is standby, not isolation.

Routing GPIO map: BCLK=47, LRCLK=48, DIN=14; OLED power=41.
Other assigned GPIOs retained; firmware changes in review/gpio-map.csv.

OLED: GPIO41 = power, GPIO11 = bus connect (both default OFF).
ON: GPIO41 high, wait for supply, GPIO11 high. OFF: GPIO11 low, then GPIO41 low.
J4 order: GND / switched 3V3 / SCL / SDA. Verify display pinout.

Front-board D1 via J6: common anode at 3V3; LOW = ON.
Wuerth 150141M173100: A=1, R=3, G=4, B=2.

ENCLOSURE INTERFACES — keyed GH cables, pin 1 to pin 1

J5: VOL- / MODE / VOL+ and hardware enable; unplugged = standby

J6: BAT button + common-anode RGB; R28-R30 remain on main board

J7: service fixture only. 3v3 is TARGET REFERENCE, not a power input.

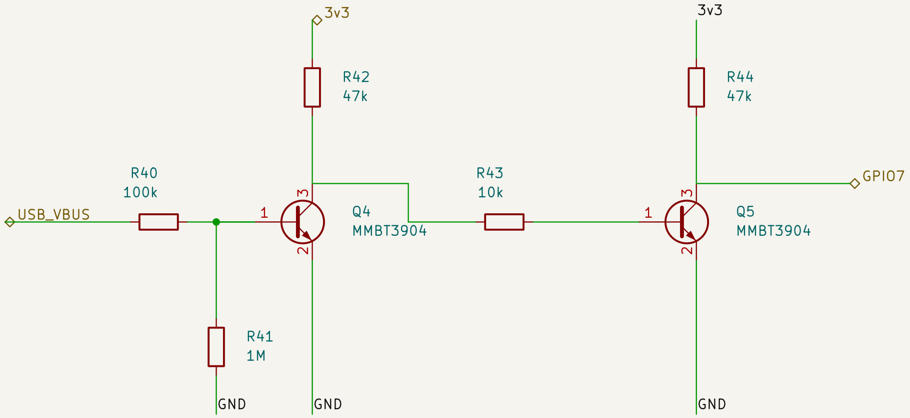
UART: TX goes to programmer RX; RX to programmer TX. Use 3.3 V logic.

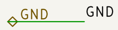
OFF is standby: protected battery still feeds regulator VIN. No hard disconnect.

Sheet: /
File: alec-main.kicad_sch
Title: ALEC main PCB
Size: A2 | Date: 2026-09-06 | **Rev: v4.2 prototype**
KiCad E.D.A. 10.0.1 | Id: 1/2

USB VBUS presence — HIGH only with a host supply connected

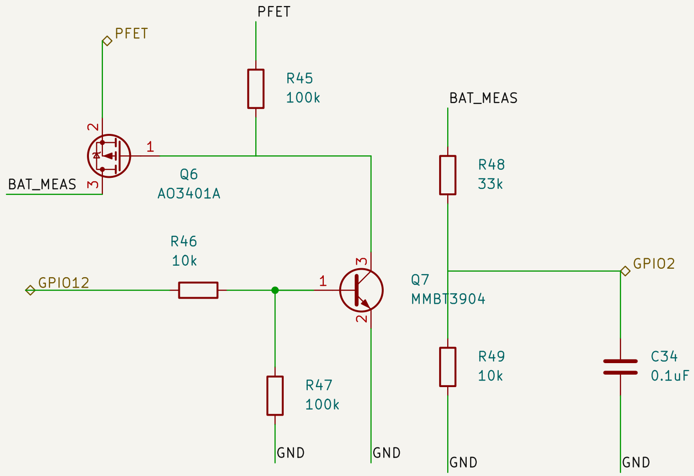
Two NPN stages keep VBUS away from the GPIO and preserve positive logic.
GPIO7: TinyUSB self_powered=true, vbus_monitor_io=7.
This is cable presence detection, not a precision USB undervoltage comparator.





Switched battery measurement — protected battery node to ADC1 / GPIO2

GPIO12 HIGH enables measurement; wait 10 ms, read calibrated ADC, then drive LOW.
Scale = 4.3; 33k / 10k, 0.1%; 100 nF filter, time constant 0.77 ms.
Designed measurement range 0-6 V; battery chemistry and percentage thresholds remain firmware settings.



Sheet: /Power monitoring/
File: power-monitor.kicad_sch

Title:

Size: A4 Date: **Rev:**
KiCad E.D.A. 10.0.1 Id: 2/2